

Top reasons to buy the Azur 851D

Our flagship DAC and digital pre-amplifier delivers exceptional quality audio. Here are 10 reasons to choose the Cambridge Audio Azur 851D over other systems...

- 1 24-bit/384kHz audio upsampling.** The 851D features ATF2 (2nd generation Adaptive Time Filtering) audio upsampling. The ATF2 system intelligently upsamples all incoming audio to 24-bit/384kHz and dramatically reduces digital jitter.
- 2 High quality digital pre-amplifier.** Thanks to the highest quality of digital preamplifiers, without bit-reduction, you can connect the 851D directly to a power amplifier such as the Azur 851W to create a truly astonishing digital music system!
- 3 Twin Analog Devices DACs.** The 851D features twin Analog Devices 1955 DACs (the Rolls Royce of DACs!) used in 'dual differential' mode for the very best digital to analogue conversion.
- 4 Toroidal transformer.** The 851D uses a large toroidal transformer to deliver perfect, low distortion power for all your music.
- 5 High performance USB.** The 851D features a high quality asynchronous USB Audio input for playback of up to 24-bit/192kHz high resolution files from a connected computer.
- 6 Superior connectivity.** Six digital inputs include S/PDIF coaxial, Toslink optical, as well as BNC coaxial and AES/EBU. Plus there's asynchronous USB Audio and Bluetooth wireless connectivity, meaning that wherever your music comes from, you can play it back in stunning quality through the 851D.
- 7 Bluetooth connectivity.** As well as the Standard Bluetooth Codec (SBC), the 851D also supports the higher quality Bluetooth system, aptX, which is supported by more and more devices and sounds even better, meaning you can get the best possible results from any Bluetooth device.
- 8 Selectable digital filters.** The 851D has three selectable filters (linear, minimum and steep) which allow you to choose the sound that best suits your ears.
- 9 High quality headphone output.** The 851D's inbuilt high-end, low distortion headphone amp, allows you to get the best possible performance from your headphones; great for when not wanting to disturb others.
- 10 Full Metal Jacket.** All the above points would mean nothing if the 851D was not built to last! From the all-metal casework and ultra rigid chassis, to the high-quality, low distortion toroidal transformer, every component in the 851D is designed to give many years of listening pleasure.

Did you know?

By using a digital pre-amplifier, the audio signal remains in the digital domain for much longer, compared to a traditional analogue integrated or pre-amplifier. Because the audio signal remains digital, and is only converted to analogue at the very last minute, it is far less susceptible to analogue signal deterioration. This means that the sound you hear is as close as possible to the original recording.

Hot Tip!

User friendly display: Features such as changing input names on the display allow you to customise the 851D to suit your system, making it simple to select and listen to sources.

Vital statistics:

DAC:	2 x Analog Devices AD1955
Digital audio inputs:	2 x S/PDIF, 4 x Toslink optical, 1 x BNC, 1 x AES/EBU, 1 x BT100 Bluetooth, 1 x USB Audio
Analogue audio outputs:	1 x Balanced XLR 1 x Unbalanced RCA 1 x headphone output
Dimensions:	115 x 430 x 360mm (4.5 x 16.9 x 17.7")
Weight:	7.5kg (16.5lbs)

Audio input signals supported

USB Audio:	16-24 bit, 32-192kHz
Toslink:	16-24 bit, 32-96kHz
S/PDIF:	16-24 bit, 32-192kHz
BNC:	16-24 bit, 32-192kHz
AES/EBU:	16-24 bit, 32-192kHz

Top-Tips



Draw every last detail from digitally stored music

The Azur 851D is our flagship DAC (Digital to Analogue Converter) and digital pre-amplifier in one powerful hub. You can play your music back from just about anywhere – from home networks, hard drives and computers, to online streaming services such as Spotify, Pandora and even YouTube. However, reproducing this digital music as the original artist intended is a complex challenge. For example, a laptop is designed to be a laptop, not a piece of hi-fi, so sound quality is usually compromised as a result.

Part of the flagship 8 Series, the 851D is for those who want to store and play their music digitally, but with no compromise on sound quality. By connecting all of your digital audio sources to the 851D, you bypass the inferior internal DACs found in the digital source device, such as a PC or laptop. The 851D then performs the critical digital to analogue conversion in far higher quality

than the source device ever could, and sends this higher quality analogue music signal on to your amplifier.

The 851D is also supplied with the BT100 Bluetooth receiver, meaning that you can wirelessly stream music direct from any Bluetooth enabled device (iPhone, iPad, smart phone laptop, Macbook) to the 851D. This extra functionality allows you to browse and control the endless content these devices offer, from the comfort of your chair.

The 851D can also be used as a digital pre-amplifier. When the optional pre-amplifier mode is enabled, the 851D features a high quality volume control, allowing you to connect directly to a power amplifier such as the Azur 851W, without the need to first connect to a traditional analogue preamplifier.

The inside story

Helping us to achieve this incredible sound are some other key features rarely seen at this price point.

- **Twin Analog Devices DACs.** The 851D features twin Analog Devices 1955 DACs (the Rolls Royce of DACs!) used in 'dual differential' mode for the very best digital to analogue conversion. Each stereo channel processes information completely separately for greater accuracy, bringing you far closer to the original recording for a truly amazing experience.

- **High performance USB.** The 851D features a high quality asynchronous USB Audio input for playback of up to 24-bit/192kHz high resolution files from a connected computer. An asynchronous USB audio input means that the 851D acts as a high quality external soundcard, allowing bit-perfect transfer from the connected computer to the 851D, ensuring all data is delivered in perfect timing for the highest possible sound quality.

- **24-bit/384kHz audio upsampling.** The 851D features ATF2 (2nd generation Adaptive Time Filtering), audio upsampling. The ATF2 system intelligently upsamples all incoming audio to 24-bit/384kHz. The ATF2 system dramatically reduces digital jitter, a form of audible interference which affects all digital music. Jitter is most apparent when music is stored on computer hard drives and is the audio equivalent to a blurred image in a movie. Data scrambling circuitry also maintains signal quality and purity during quieter periods by reducing background noise.

Did you know?

As well as the Standard Bluetooth Codec (SBC), the 851D also supports the higher quality Bluetooth system, aptX, which is supported by more and more devices and sounds even better, meaning you can get the best possible results from your Bluetooth device.

Toroidal vs EI transformers

We strive for audio perfection which is why the 851D features an audiophile grade toroidal transformer. Cambridge Audio was one of the first manufacturers to use toroidal transformers in the 1970s, and we continue to use them today. Some of the benefits of a toroidal transformer, over the cheaper, inferior EI transformers used by most other manufacturers are:

- ✓ High quality power output to the amplification circuits, with no clipping of the waveform signal, to provide a consistent signal
- ✓ No transformer noise, buzz or hum
- ✓ Large power reserves, to allow the transformer to deliver high peaks of volume when required
- ✓ Fully shielded against internal and external electrical interference, to maintain quality of power and signal



A typical cheaper EI transformer:

- ✗ Weak inconsistent signal
- ✗ No shielding
- ✗ Noisy
- ✗ Won't drive powerful and demanding speakers

In a different class!

The 851D features an asynchronous USB audio input, acting as a high quality external soundcard, allowing you to send up to 24-bit/192kHz high resolution files to the 851D from your computer.

- **High resolution audio** - most computers can't play high resolution audio through their headphone output. Featuring asynchronous USB audio clocking, and supporting both USB Class 1 and USB Class 2, the 851D can be used to play everything from MP3s to the highest quality 24-bit/192kHz studio master recordings from your computer, all free from unwanted noise, jitter and distortion.
- **Class 1 and Class 2 audio** - The 851D can be switched into USB Class 2 mode, allowing you to play everything from MP3 to the highest quality 24-bit/192kHz studio master recordings straight from your computer.
- **Dedicated USB driver** - when running in Class 2 mode, the 851D can also use Cambridge Audio's dedicated Windows USB Class 2 USB driver. This allows you to bypass the computer operating system's poor audio path, preventing unwanted interference and re-sampling, guaranteeing a professional quality audio output.

BT100 Bluetooth input

Asynchronous USB Audio input



RS232 control for custom installation integration

Digital loop outputs

Did you know?

The 851D has three selectable filters (linear, minimum and steep) which allow you to choose the sound that best suits your ears, and also features an inbuilt high-end, low distortion headphone amp, allowing you to get the best possible performance, however you choose to listen.

Superior connectivity

Six digital inputs include S/PDIF coaxial, Toslink optical, as well as BNC coaxial and AES/EBU. Plus there's asynchronous USB Audio and Bluetooth wireless connectivity, meaning that wherever your music comes from, you can play it back in stunning quality through the 851D. All inputs are upsampled to 24-bit/384kHz output. This means you can connect a world of digital source components from set-top boxes to media streamers, smartphone docks to computers, and whatever you connect, the results are astonishing.

The 851D features both unbalanced RCA and balanced XLR audio outputs, as found in many modern high-end hi-fi separates such as our 851E and 851W pre and power amplifiers.

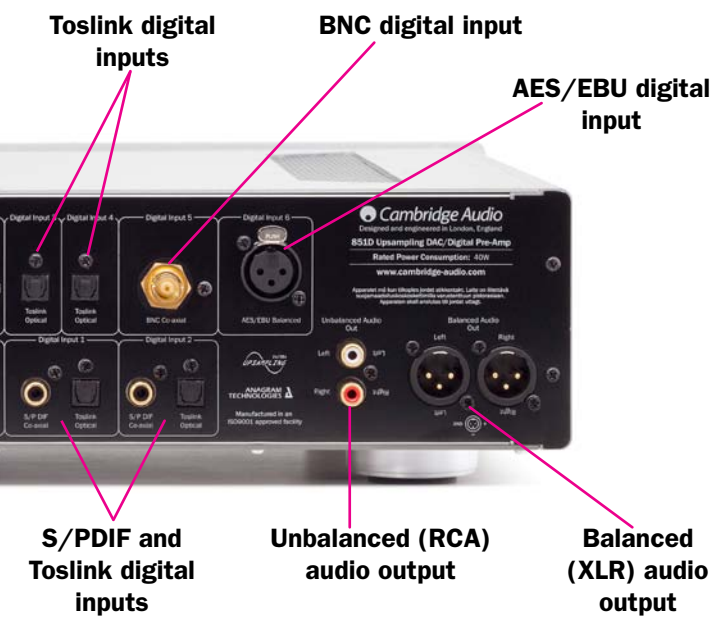
Designer's note



"Listen to the ATF2 upsampling in the 851D applied to a 'lossless' digital file and you can hear every detail, every bit of emotion you'd hear from vinyl. This is the future of music."
[Sam, Product Manager]

Azur 851D

Upsampling DAC and Digital Pre-amplifier



Digital Pre, with Analogue Power!

Why not connect the 851D directly to the Azur 851W power amplifier, to create a truly astonishing digital music system!

Unlike many digital pre-amplifiers, the 851D's digital pre-amplifier is the highest possible quality thanks to the powerful 32-bit Blackfin DSP (Digital Signal Processor), which we also use to drive our ATF audio upsampling system. Most digital preamplifiers use 'bit reduction' to adjust the level of the volume output, however this results in much lower sound quality because digital information is being cut out of the original audio signal.

Instead, the 851D digital pre-amplifier uses the Blackfin DSP to control the level of the volume output, meaning that the entire digital signal remains fully intact for the best possible sound.

Maximise streaming services and Apps

Bring the 851D's stunning, superior audio quality to all your apps and streaming services via your smartphone, tablet or computer. Instantly and easily listen to popular services such as Last.fm, Spotify or BBC iPlayer Radio played back in stunning quality through the 851D's Bluetooth connection.

All metal design

Electronics (and therefore sound quality) are particularly susceptible to vibration; that's why we've used a low resonance, acoustically-dampened metal chassis to eliminate as much vibration as possible; ensuring that the highest sound quality is extracted from each audio source.

